

DIY RC FOAM AIRPLANE BUILD

Technical Documentation, Component Schematics & Transmitter Configuration

Aircraft Type:	4-Channel High-Wing Trainer	Radio Protocol:	FLYSKY FS-i6X (2.4GHz AFHDS 2A)
Document Ref:	DOC-RCPLANE-1400KV-01	Power System:	1400KV BLDC + 3S LiPo + 30A ESC

1. AIRCRAFT OVERVIEW & TECHNICAL SPECIFICATIONS

This document details the build specifications, electronics setup, power system parameters, and transmitter channel mapping for the custom **DIY High-Wing Foam RC Plane**. Designed as a stable trainer aircraft, the high-wing setup offers natural self-righting tendencies, combined with full 4-channel control (Aileron, Elevator, Rudder, and Throttle).

Component / Parameter	Specification	Operational Notes
Brushless Motor	2212 1400KV Outrunner	Optimal prop: 8x4.5 or 9x4.7 2-blade
Electronic Speed Control	30A ESC with 5V/2A BEC	Powers receiver and servos via internal BEC
Battery Pack	3S LiPo (11.1V / 1000mAh – 2200mAh)	Discharge Rate: 25C to 35C
Transmitter / Receiver	FlySky FS-i6X TX & FS-iA6B / FS-A8S RX	2.4GHz 6 to 10 Channel AFHDS 2A
Control Servos	4x SG90 Micro Servos (9g)	Aileron Left, Aileron Right, Elevator, Rudder
Airframe Material	Depron / EPS Foam & Kraft Paper reinforcement	Lightweight High-Wing Trainer configuration

2. RECEIVER PIN MATRIX & CHANNEL MAPPING (FLYSKY FS-I6X)

The FlySky FS-i6X transmitter uses standard Mode 2 channel assignments. Connect each control surface servo and the ESC signal lead to the corresponding receiver channel:

Receiver Channel	Control Surface / Function	Connected Component	Movement & Trimming Direction
Channel 1 (CH1)	Aileron (Roll Control)	2x SG90 Servos (via Y-Harness)	Right Stick Left/Right. Differential movement for banking turns.
Channel 2 (CH2)	Elevator (Pitch Control)	1x SG90 Servo (Horizontal Tail)	Right Stick Up/Down. Up elevator raises tail flap (Nose UP).
Channel 3 (CH3)	Throttle (Speed Control)	30A ESC Signal Lead (3-pin BEC)	Left Stick Up/Down. Powers motor & supplies 5V to receiver.
Channel 4 (CH4)	Rudder (Yaw Control)	1x SG90 Servo (Vertical Tail)	Left Stick Left/Right. Controls directional yaw on tail.
Channel 5 / 6	Auxiliary (Optional)		

RECEIVER CHANNEL	CONTROL SURFACE / FUNCTION	CONNECTED COMPONENT	MOVEMENT & TRIMMING DIRECTION
		Flaps / Flight Controller / Dual Aileron	Assignable to SwA, SwC switches on FS-i6X.

SAFETY PRECAUTION: ESC CALIBRATION & PROPELLER HAZARD

NEVER install the propeller onto the 1400KV motor while setting up the receiver or performing ESC throttle calibration! A sudden motor start can cause severe injuries or slice through foam structures.

3. BILL OF MATERIALS (BOM)

ITEM	COMPONENT NAME	SPECIFICATIONS	QTY
1	2212 1400KV Brushless Motor	Outrunner BLDC Motor + Prop Saver/Adapter	1
2	30A Brushless ESC	30A Continuous, 5V/2A BEC, T-Plug/XT60	1
3	3S LiPo Battery	11.1V 1500mAh–2200mAh 30C	1
4	FlySky FS-i6X + Receiver	2.4GHz Transmitter & 6-Ch Receiver	1 Set
5	TowerPro SG90 / SG90s Servos	9g Micro Nylon/Metal Gear Servos	4
6	Propeller	8x4.5 or 9x4.7 Electric Fly Propeller	1 or 2
7	Servo Y-Harness & Extensions	15cm Servo Y-Cable for Ailerons	1
8	Pushrods & Control Horns	1.2mm Steel Wire + Plastic Control Horns	4 Sets
9	Airframe Construction	Depron/Thermocol Foam, Hot Glue, Reinforced Tape	1 Airframe

4. ESC THROTTLE CALIBRATION PROCEDURE

1. Remove the propeller from the motor shaft.
2. Turn ON the FlySky FS-i6X transmitter and push the throttle stick (Left Stick) all the way to **MAX (100%)**.
3. Connect the 3S LiPo battery to the 30A ESC.
4. The ESC will play a musical chime followed by **2 short beeps** (indicating MAX throttle point captured).
5. Immediately lower the throttle stick all the way to **MIN (0%)**.
6. The ESC will play a confirmation chime followed by 3 beeps (for 3S battery cells) and 1 long arming beep.
7. ESC calibration is complete! Throttle control is now fully operational.

5. PRE-FLIGHT BALANCE & CENTER OF GRAVITY (CG) CHECKLIST

Finding the Center of Gravity (CG)

For a high-wing trainer, the ideal Center of Gravity (CG) is located at **25% to 30% of the wing chord** measured back from the leading edge of the main wing. Shift your 3S LiPo battery forward or backward in the fuselage compartment until the plane balances level on two fingertips under the wing.

- **Control Surface Check:** Verify Ailerons move opposite to each other (Right roll = Right aileron UP, Left aileron DOWN).
- **Motor Rotation:** Motor must rotate counter-clockwise (viewed from front) so the propeller pushes air backward over the wing. If it rotates in reverse, swap any two of the three bullet connector wires between the ESC and motor.
- **Range Test:** Conduct a transmitter range test before the maiden flight.